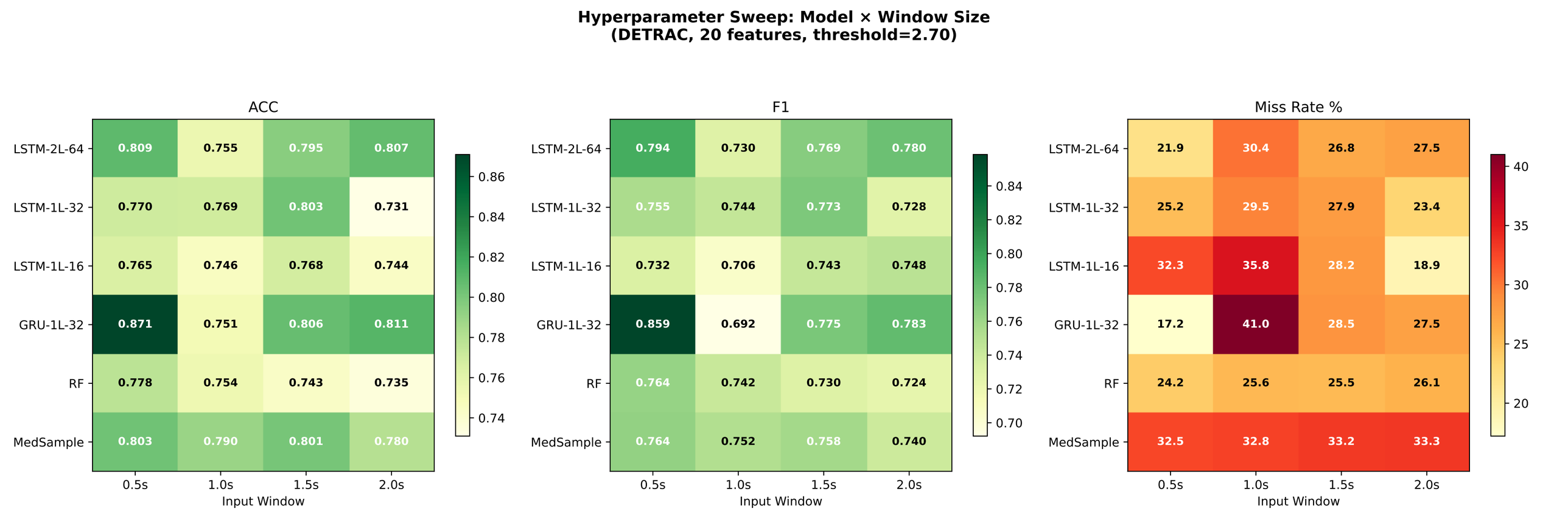
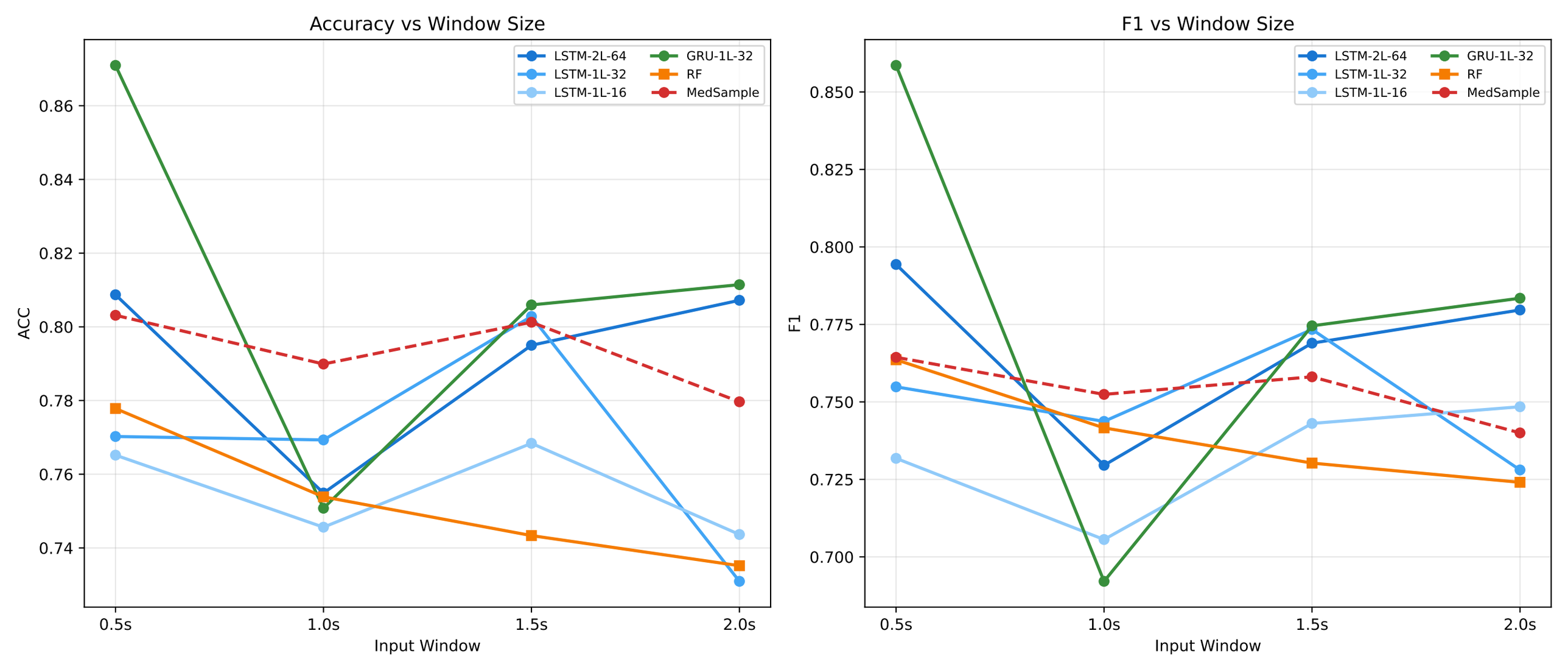

Hyperparameter Sweep: Model × Window Size
(DETRAC, 20 features, threshold=2.70)
ACC
LSTM-2L-64
LSTM-1L-32
LSTM-1L-16
GRU-1L-32
RF
MedSample
0.809 0.755 0.795 0.807
0.770 0.769 0.803 0.731
0.765 0.746 0.768 0.744
0.871 0.751 0.806 0.811
0.778 0.754 0.743 0.735
0.803 0.790 0.801 0.780
0.86
0.84
0.82
0.80
0.78
0.76
0.74
0.5s
1.0s
1.5s
2.0s
Input Window
F1
LSTM-2L-64
LSTM-1L-32
LSTM-1L-16
GRU-1L-32
RF
MedSample
0.794 0.730 0.769 0.780
0.755 0.744 0.773 0.728
0.732 0.706 0.743 0.748
0.859 0.692 0.775 0.783
0.764 0.742 0.730 0.724
0.764 0.752 0.758 0.740
0.84
0.82
0.80
0.78
0.76
0.74
0.72
0.70
0.5s
1.0s
1.5s
2.0s
Input Window
Miss Rate %
LSTM-2L-64
LSTM-1L-32
LSTM-1L-16
GRU-1L-32
RF
MedSample
21.9 30.4 26.8 27.5
25.2 29.5 27.9 23.4
32.3 35.8 28.2 18.9
17.2 41.0 28.5 27.5
24.2 25.6 25.5 26.1
32.5 32.8 33.2 33.3
40
35
30
25
20
0.5s
1.0s
1.5s
2.0s
Input Window


Accuracy vs Window Size
ACC
0.86
0.84
0.82
0.80
0.78
0.76
0.74
0.5s
1.0s
1.5s
2.0s
Input Window
LSTM-2L-64
LSTM-1L-32
LSTM-1L-16
GRU-1L-32
RF
MedSample
F1 vs Window Size
F1
0.850
0.825
0.800
0.775
0.750
0.725
0.700
0.5s
1.0s
1.5s
2.0s
Input Window
LSTM-2L-64
LSTM-1L-32
LSTM-1L-16
GRU-1L-32
RF
MedSample

Hyperparameter Sweep — Full Results

Window	Model	Params	Epoch	Acc	F1	AUC	Miss%	Mixed Med
0.5s	MedSample	0	-	0.803	0.764	—	32.5%	—
0.5s	RF	-	-	0.778	0.764	0.844	24.2%	0.815
0.5s	LSTM-2L-64	57,442	46	0.809	0.794	0.901	21.9%	0.838
0.5s	LSTM-1L-32	7,474	4	0.770	0.755	0.854	25.2%	0.857
0.5s	LSTM-1L-16	2,738	4	0.765	0.732	0.812	32.3%	0.811
0.5s	GRU-1L-32	5,746	9	0.871	0.859	0.919	17.2%	0.865
1.0s	MedSample	0	-	0.790	0.752	—	32.8%	—
1.0s	RF	-	-	0.754	0.742	0.824	25.6%	0.756
1.0s	LSTM-2L-64	57,442	7	0.755	0.730	0.796	30.4%	0.769
1.0s	LSTM-1L-32	7,474	18	0.769	0.744	0.822	29.5%	0.776
1.0s	LSTM-1L-16	2,738	7	0.746	0.706	0.772	35.8%	0.721
1.0s	GRU-1L-32	5,746	8	0.751	0.692	0.796	41.0%	0.800
1.5s	MedSample	0	-	0.801	0.758	—	33.2%	—
1.5s	RF	-	-	0.743	0.730	0.816	25.5%	0.741
1.5s	LSTM-2L-64	57,442	20	0.795	0.769	0.875	26.8%	0.864
1.5s	LSTM-1L-32	7,474	28	0.803	0.773	0.843	27.9%	0.842
1.5s	LSTM-1L-16	2,738	10	0.768	0.743	0.834	28.2%	0.750
1.5s	GRU-1L-32	5,746	17	0.806	0.775	0.821	28.5%	0.864
2.0s	MedSample	0	-	0.780	0.740	—	33.3%	—
2.0s	RF	-	-	0.735	0.724	0.807	26.1%	0.643
2.0s	LSTM-2L-64	57,442	15	0.807	0.780	0.869	27.5%	0.850
2.0s	LSTM-1L-32	7,474	0	0.731	0.728	0.780	23.4%	0.800
2.0s	LSTM-1L-16	2,738	2	0.744	0.748	0.806	18.9%	0.812
2.0s	GRU-1L-32	5,746	19	0.811	0.783	0.814	27.5%	0.833